

RAMCO INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
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Degree, Semester & Branch: III Semester B.E. Computer Science and Engineering

Course Code & Title: CS8391 Data Structures

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Innovative practice: Role-Play

Topic: Sorting Techniques- Selection Sort, Merge Sort

Objectives of the activity:

- To enable the students to understand the concepts of sorting technique and improve the communication skills, teamwork.

Justification for choosing the topic:

Different types of sorting techniques are available, each having a unique strategy for arranging the elements. In order to differentiate the types of sorting methodology, the role-play activity is suitable for them to remember the working of sorting.

Description:

Role play is an interesting activity which helps the students to naturally develop and use their cognitive abilities and skills. It encourages creativity and imagination among students. Role-play is a technique that allows the students to explore realistic situations by interacting with other people in a managed way in order to develop experience and trial different strategies in a supported environment.

Strategy for Team Formation:

Two homogeneous teams were formed. One team consist of boys students and another team comprises the girl students of the class.

Activity:

The activity was conducted in the classroom in the sorting techniques selection sort and merge sort. Two teams of students were involved and demonstrated how the sorting works. From the first team one student explained the working process of selection sort and the remaining members acted according to that as shown in Figure 1.

The second team demonstrated the working of merge sort which is based on divide and conquer strategy. First, the team members are divided into two groups based on the middle value. In each group the same procedure is applied until it become a group contains single person. After that, they started the merging process. The members present in the two nearby groups are compared based on the numbers, arranged in order among themselves and

form a single group. This process was repeated by groups until all the students came into single group. This role play is depicted in Figure 2.

Observations made:

Based on their performance in role play the following observations were made.

- Students can easily visualize the working of sorting technique and how the swapping performs.
- Some students need more clarification on the execution of merge sort.

Actions taken:

- Additional problems were discussed to make them to familiarize the concept.

Outcome:

Students were actively involved in this role play. This activity gave a better understanding and the visualization helped them to remember the working process of various sorting methods.

Relevance Mapping - This activity helps in attaining the following Objectives – PO mapping:

Objectives / PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
O1	3	3	3	3	-	-	-	-	3	3	-	1

Reflective Report

Pre-implementation:

Identified Challenges:

- Few students do not understand the concept clearly and made a mistake
- In selection sorting, after comparing the first element with the next element if smaller element is found then they immediately swapped with other instead of finding smallest element from all elements and swapping.
- Some of the students feel shy to participate in this activity.

Addressing challenges

- I explained the sorting concepts once again to make them clear.
- The students are motivated to actively participate in this activity.

Post implementation

- All the students enjoyed this type of activity.
- Students can easily remember the sorting procedure.



Figure 1: Selection Sort Role play



Figure 2: Merge Sort Role play

References:

1. Mr.R.Venkatesh, "Caesar Cipher" in course "CS6701 Cryptography and Network Security" dated:08.07.2019 <https://www.ritrjpm.ac.in/images/computer-science/Roleplay.pdf>
2. Mrs.M.SwarnaSudha, "Swapping of two number" in course "GE8151 Problem Solving and Python Programming" dated:17.08.2019.
(<https://www.ritrjpm.ac.in/images/computer-science/Roleplay-MSS1.pdf>)

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